

Session 3.6

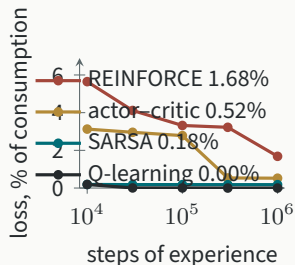
The Growth Model Across Algorithms

The full table Deck 3.2 opened; then actor-critic on the continuous model, and the bridge to Session 4



Zhigang Feng

Every Method of the Session, on One Model and One Grade

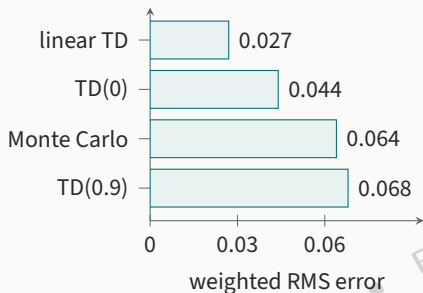


Method	Estimates	Loss: weighted / all
MC, exploring starts	Q^g	- / 0.20%
SARSA	best ϵ -soft rule's value	0.18% / 0.38%
Q-learning	Q^*	0.00% / 0.01%
REINFORCE	the parameters θ	1.68% / 2.10%
actor-critic	θ and V^{g_θ}	0.52% / 1.24%

- **One instance, every row and every curve:** the recorded **tabular** run on the 7×2 grid — 14 states, 20 seeds, 10^6 steps. Four of the five are on the figure.
- **Replaced:** the expectation by samples, the table by parameters, the max by a gradient step.

Source: Recorded by the book's script; tabular, 7×2 grid, 10^6 steps (MC: 1.8×10^6). Miao & Feng, ch. 12.

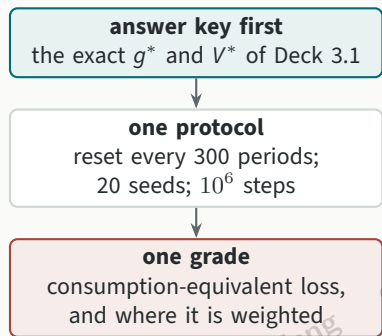
Five Readings You Could Not Make in Deck 3.2



- ▶ **Sampling leads early, bootstrapping late:** at 10^3 steps Monte Carlo is at 0.596, TD(0) at 1.015.
- ▶ **Monte Carlo control needs exploring starts** — greedy covers 3 cells of 68 and loses 2.73%. **SARSA's 0.18% is not an error:** it is the best ϵ -soft rule's own value.
- ▶ **The critic is the baseline:** actor-critic 0.52% against REINFORCE's 1.68%.
- ▶ **Deck 3.2's demo is on none of these rows** — continuous model, 0.014%.

Source: Recorded by the book's script for ch. 12, tabular, 7×2 grid, $\beta = 0.95$; evaluation errors weighted by $\bar{d}$, medians of 20 paths. Bars at 10^5 steps, except the linear fixed point. Last bullet: the continuous demo of Deck 3.2.

The Protocol Is the Part That Travels



- ▶ Every number is a distance from a benchmark built before any learner ran.
- ▶ **Say where the grade is taken, and on what.** Tabular, 7×2 : 0.52% where the economy lives, 1.24% over all 14 states. The continuous demo: 0.014% and 0.015%.
- ▶ **Report medians and seed counts.** That run's best seed is 0.35%, worst 2.67%; the demo's, 0.002% and 0.106%.
- ▶ **Session 4 has no closed form: its answer key must be built classically first.**

Source: Tabular run recorded by the book's script, 7×2 grid, 20 seeds; the continuous demo from Session 3 lab, block A2.

Session 3.6

Deep Value Iteration

Actor-critic with a known model, on Session 2's continuous growth model



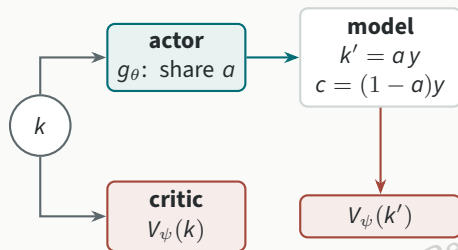
From the Grid to a Network

	Value function iteration	Deep value iteration
states	a fixed grid	a fresh sample every step
value	a table, interpolated	a network $V_{\psi}(k)$
improvement	exact $\max_{k'}$ at each point	a policy network, by gradient ascent
cost in dimension d	N^d	grows with the sample
convergence	a contraction	no general guarantee

- ▶ The capital stock is now continuous: no table of states, no list of actions to compare.
- ▶ **The trade: give up the exact $\max$ and the contraction; gain reach in dimension.**

Source: Maliar, Maliar & Winant, *JME* 122, 2021; Miao & Feng, *AI for Economists*, ch. 12, “When DP, when RL”.

Two Networks: an Actor and a Critic, with the Model Known



the actor's gradient passes
through the model and the critic

- ▶ **Actor**: Session 2's saving-share network.
Critic: a value network.
- ▶ The model is known and differentiable: the actor takes a **pathwise gradient** of $u(c) + \beta V_\psi(k')$ through the budget constraint.
- ▶ Deck 3.5's score-function gradient was for unknown models. This one has far less variance.
- ▶ **Same two roles as Deck 3.5. Only the source of the actor's gradient changed.**

Source: Miao & Feng, *AI for Economists*, ch. 12, "Actor-critic for the household problem"; Session 3 lab, block F.

The Value Step, and Why the Target Is Detached

- ▶ Hold the actor fixed. One Bellman backup under it is the label:

$$y(k) = u(c_{\theta}(k)) + \beta V_{\psi}(k'_{\theta}(k)).$$

- ▶ Fit the critic by least squares on sampled states: $\min_{\psi} \frac{1}{n} \sum_i (V_{\psi}(k_i) - y(k_i))^2$. Session 2's supervised learning, with a Bellman target as the label.
- ▶ **Detach** the target: no gradient flows through y . Otherwise the network lowers the loss by moving the label towards the prediction.
- ▶ An exact backup shrinks the value error by β . A fitted backup adds a regression error, every round.
- ▶ **The network being trained produces its own label: bootstrapping, as in TD(0), without sampling noise.**

Source: Miao & Feng, *AI for Economists*, ch. 12, “Neural networks and DQN”; V. Mnih et al., *Nature* 518, 2015.

The Policy Step: Climb the Right-Hand Side of the Bellman Equation

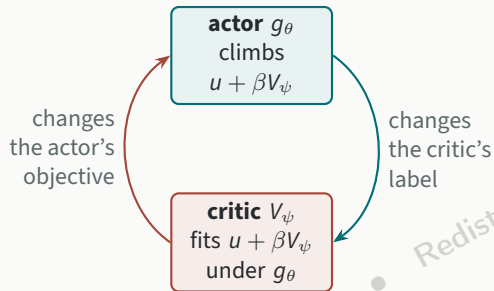
- ▶ Hold the critic fixed. Improve the actor by gradient *ascent* on

$$\max_{\theta} \frac{1}{n} \sum_i \left[u(y_i - k'_{\theta}(k_i)) + \beta V_{\psi}(k'_{\theta}(k_i)) \right], \quad y_i = Ak_i^{\alpha} + (1 - \delta)k_i.$$

- ▶ A grid solver compares every k' . Here a gradient step replaces the max; no inner optimisation is solved.
- ▶ Structure is built into the actor, not penalised: the sigmoid share keeps $c > 0$ and $k' > 0$ at every θ .
- ▶ The actor needs the critic's *slope*, $V'_{\psi}(k')$, not its level. A critic can fit levels well and slopes badly.
- ▶ **Two networks, because the max over a network's output is itself a hard problem. The actor amortises it.**

Source: Session 3 lab, block F; Session 2.5, "Feasibility by Construction".

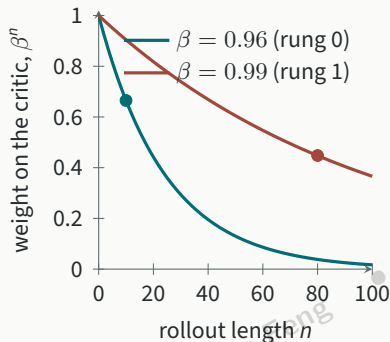
Two Moving Targets



- ▶ Each network's target depends on the other. Neither solves a fixed problem.
- ▶ The pair can agree and still be wrong: a critic that correctly values a bad actor satisfies its own Bellman equation.
- ▶ Practice: let the critic move faster than the actor; start both where the answer is known.
- ▶ **Here that is the steady state:**
 $V(k^*) = u(c^*) / (1 - \beta)$ **fixes the critic's level.**

Source: V. Konda, J. Tsitsiklis, "Actor-critic algorithms", *NeurIPS* 12, 2000; Session 3 lab, block F.

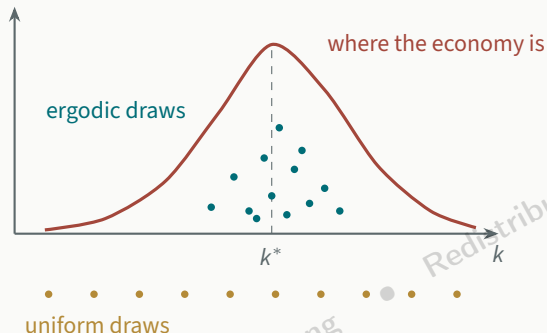
One Step or Many



- ▶ Roll the actor n periods through the known model, then ask the critic:
 $J_n(k) = \sum_{t < n} \beta^t u(c_t) + \beta^n V_\psi(k_n)$.
- ▶ $n = 1$: one-step fitted value iteration.
Larger n leans less on the critic: its weight is β^n .
- ▶ With shocks, long rollouts add variance. This model is deterministic, so they add none.
- ▶ **The lab:** $n = 10$ on rung 0, $n = 80$ on rung 1. Critic weights 0.66 and 0.45.

Source: Deck 3.4, n -step returns; Session 3 lab, block F (Session 2's rungs 0 and 1).

Where to Sample



- ▶ The sampling distribution weights the critic's regression: accuracy is bought where states are drawn.
- ▶ **Ergodic sampling:** simulate under the current actor, train where the economy goes. Efficient, and blind to rare states.
- ▶ The lab keeps Session 2's band, uniform in $\log k$ on $[0.2, 2] \times k^*$; off-domain errors are reported apart.
- ▶ **A solution is claimed only where it was trained and tested.**

Source: Schematic. Judd, Maliar & Maliar, *Quantitative Economics* 2(2), 2011; Session 2.5, "Where to Sample".

The Code: One Training Step

```
def rollout_objective(actor, critic, k, n_step):          # J_n(k)
    crra, beta = rung.gamma, rung.beta                  # curvature, discount
    total, disc, z = 0.0, 1.0, torch.ones_like(k)
    for _ in range(n_step):
        k_next = actor(k, z)                            # feasible by construction
        total = total + disc * utility(actor.resources(k, z) - k_next, crra)
        disc, k = disc * beta, k_next
    return total + disc * critic(k)
loss_actor = -rollout_objective(actor, critic, k, n).mean() # climb J
loss_actor.backward(); opt_actor.step()
with torch.no_grad():                                   # detach the label
    target = rollout_objective(actor, critic, k, n)
loss_critic = (critic(k) - target).square().mean()
loss_critic.backward(); opt_critic.step()
```

- ▶ The listing keeps Session 2's beta for the discount, crra for the curvature.
- ▶ **No benchmark enters either loss; the closed form is kept for grading.**

Source: Session 3 lab, block F; the training step of Miao & Feng, *AI for Economists*, ch. 12.

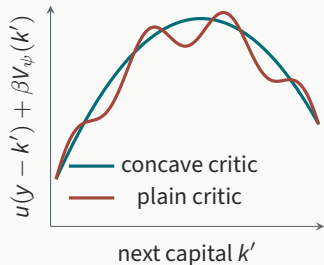
Session 3.6

Structure, and the Choice of Road

What a concave critic buys, and how to tell a right answer from a quiet residual



Value Functions Have a Shape, and a Critic Can Be Built to Keep It



- ▶ Concave utility and a convex constraint set make V increasing and **concave**; the Bellman operator preserves both.
- ▶ A plain network knows none of it, and the actor climbing $u + \beta V_\psi$ follows every bump.
- ▶ An **input-concave** critic (Session 2.6's input-convex network, negated) holds the shape at every ψ .
- ▶ **Then the actor's objective has one peak, found by gradient ascent from any start.**

Source: Schematic. Stokey, Lucas & Prescott, *Recursive Methods*, 1989, §4.2; Amos, Xu & Kolter, *ICML*, 2017; Session 2.6.

What That Buys, What It Does Not, and What It Costs

- ▶ Fitted value iteration is a backup, then a regression: $V_{n+1} = \Pi(TV_n)$, with Π the fit onto the network class.
- ▶ T is a contraction. ΠT need not be, and concavity of the class does not make it one: a projection after a contraction is still a projection. That is the “no general guarantee” row.
- ▶ What the class does buy: the label TV_n stays inside it, so the fit never bends a target into a shape it cannot take, and the actor never chases a bump the critic invented.
- ▶ The cost is measured: 7.8 times the RMSE on Session 2.6’s test. It must come from theory: if a discrete choice kinks the true value, concavity is bias.
- ▶ **This lab’s critic is unconstrained; the gates protect it. Buy structure when theory supplies it, and test as if you had not.**

Source: G. Gordon, “Stable function approximation in dynamic programming”, *ICML*, 1995; Session 2.6.

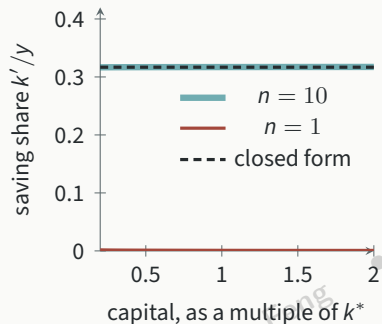
Euler or Bellman: Measured on Session 2's Rungs

held-out states	rung 0		rung 1	
	Euler	Bellman	Euler	Bellman
max policy error (gate: 1%)	0.64%	0.14%	0.04%	0.25%
max value error, % of consumption (gate: 0.1%)	–	0.07%	–	0.08%
simulated against analytical k^* (gate: 1%)	0.47%	0.12%	0.18%	0.32%
$\log_{10}$ max Euler error	–1.95	–2.78	–3.34	–2.20
seconds, one CPU core	1.4	19	4.6	165

- ▶ Euler: one network, a sharper condition, no value. Bellman: two networks, no first-order conditions needed (kinks, discrete choices), and a value for welfare comparisons.
- ▶ **A decision, not a doctrine: take the Euler road when the first-order conditions are clean.**

Source: Session 2 lab, block D; Session 3 lab, block F. Session 2 calibration: $\alpha = 0.33, A = 1$; rung 0: $\beta = 0.96, \delta = 1$, log utility; rung 1: $\beta = 0.99, \delta = 0.025$, CRRA 2.

A Small Residual Is Not a Right Answer



rung 0, same seed, same steps	$n = 10$	$n = 1$
$\log_{10}$ max Bellman residual	-1.77	-1.78
max policy error	0.14%	99.7%
steady-state gap	0.12%	91%

- ▶ With $n = 1$ the critic values a bad actor *correctly*: that pair satisfies its own Bellman equation too.
- ▶ **The residual cannot tell them apart. The benchmark and the steady state can.**

Source: Session 3 lab, block F (Session 2 rung 0); Miao & Feng, ch. 12, “Knowing whether the answer is right”.

Session 3.6

From One Household to an Economy

When to use what, and what Session 4 inherits



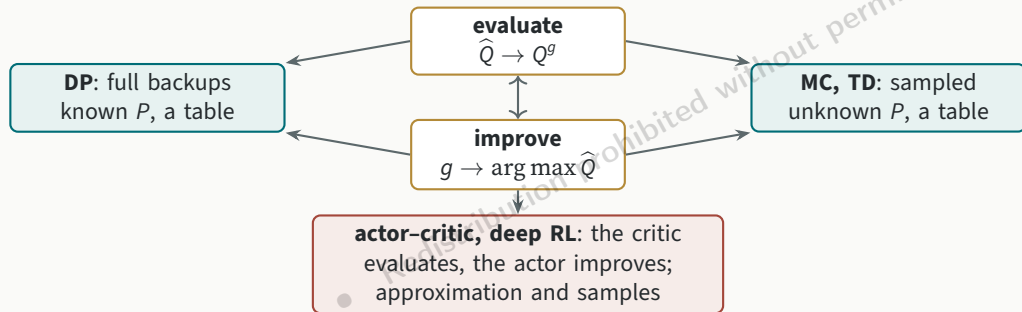
When Dynamic Programming, When Reinforcement Learning

	model known	model unknown
low dimension	DP: nothing beats it	tabular RL; coverage is the issue
high dimension	approximate DP: deep value iteration	deep RL's home ground

- ▶ Bottom left is where most structural economics lives: the model can be simulated, the grid cannot be built.
- ▶ Already in use: dynamic discrete choice, where Rust's estimation re-solves the Bellman equation inside the parameter search; high-dimensional macro; interest-rate and tax rules; and platform pricing, where the environment holds other agents' responses.
- ▶ **The caveat: reinforcement learning scales the *solution*, not the *identification*. The data discipline lives in the calibration.**

Source: J. Rust, *Econometrica* 55(5), 1987; Hinterlang & Tänzer, Bundesbank DP 51/2021; Miao & Feng, ch. 12, "When DP, when RL".

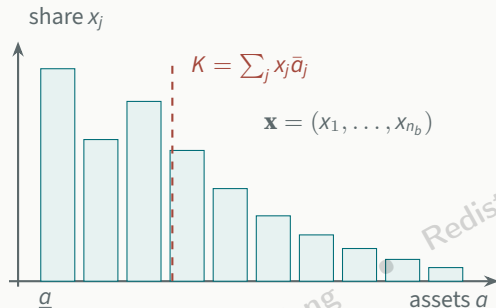
One Backbone: Evaluate, Improve



- ▶ VFI: explicit sums, pure greedy. MC control: sample means, ϵ -greedy. SARSA, Q-learning: bootstrapped samples. Actor-critic: both steps on every transition.
- ▶ **Deck 3.2 opened with this loop and one method on it. Here it is with every method of the session attached.**

Source: After Miao & Feng, *AI for Economists*, ch. 12 (the algorithmic backbone); Sutton & Barto, 2018, §4.6.

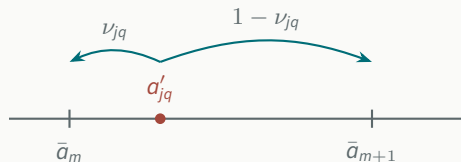
Bridge (1): Put the Distribution in the State, as a Histogram



- ▶ The method starts by choosing what to carry of the wealth distribution. Krusell and Smith carry its mean.
- ▶ Carry a **histogram** instead: shares x_j in n_b asset bins. No shape assumed; the borrowing-limit mass and the tail survive.
- ▶ Prices follow inside the model: K , then r and w from the firm's conditions.
- ▶ **The state is $n_b + 2$ numbers, which a network takes as one input vector.**

Source: Schematic after Miao & Feng, ch. 12; method of Feng, Han, Sargent & Zhu (2026, working paper), research in progress.

Bridge (2): an Analytic Transition Matrix, Not a Simulated Panel



mass is split between neighbouring bins in proportion to distance

$$\mathbf{x}_{t+1} = \mathbf{x}_t \hat{\mathbf{M}}_g(\mathbf{x}_t)$$

- ▶ Households in bin j with shock node z_q choose a'_{jq} ; its mass is split between the two neighbouring bins, and summing over nodes gives a transition matrix $\hat{\mathbf{M}}_g$. Mass is conserved exactly.
- ▶ The split is piecewise linear in a' , so $\mathbf{x}_{t+1}$ is *differentiable* in the policy's parameters. A simulated panel is not.
- ▶ **This is the discrete-time answer to “ Γ_g has no closed form” that Deck 3.1 promised.**

Source: The two-point split is E. Young, *JEDC* 34(1), 2010. Using it as an analytic, differentiable law of motion a learner can train against is Z. Feng, J. Han, T. Sargent & S. Zhu (2026, working paper). Miao & Feng, ch. 12, “An analytic transition matrix”.

Bridge (3): the Household's Actor-Critic

Household problem, no aggregate shocks

- 1 initialise critic $V_\psi(a, z, \mathbf{x})$, actor $a' = g_\theta(a, z, \mathbf{x})$, target copy $\psi^- \leftarrow \psi$
- 2 **repeat:** draw a batch of distributions $\mathbf{x}$ (perturbed, and pushed forward under the actor)
- 3 prices: $K = \sum_j x_j \bar{a}_j$, then r, w ; choices a' and c at every $(\bar{a}_j, z_q)$
- 4 $\mathbf{x}' = \mathbf{x} \hat{\mathbf{M}}_g$, **held constant when differentiating**
- 5 critic step: regress on $y = u(c) + \gamma \sum_q p_q V_{\psi^-}(a', z_q, \mathbf{x}')$
- 6 actor step: pathwise gradient of $u(c) + \gamma \sum_q p_q V_\psi(a', z_q, \mathbf{x}')$; sync ψ^- at intervals
- 7 **until** Euler residuals are small *and* forward-simulated aggregates stop moving
 - Block F with a longer input. Line 4 is the economics: an atomistic household does not count its own effect on $\mathbf{x}'$.
 - **A method, shown without results: the research is in progress. Session 4 owns equilibrium, the government, and validation.**

Source: Miao & Feng, *AI for Economists*, ch. 12; Feng, Han, Sargent & Zhu (2026, working paper), research in progress.

The Six Things Deck 3.2 Used on Credit, and What Each One Cost

The idea, used on credit	Paid	What the answer cost on this model
sample the expectation	3.3	greedy covers 3 cells of 68 and loses 2.73%; exploring starts, 0.20%
bootstrap one step	3.4	the step size alone moves TD(0) from 1.270 to 0.044
approximate the value	3.5	three bases: same weighted error, maximum error 14 times apart
improve by a gradient	3.5	tabular, 7×2 : REINFORCE 1.68%, actor-critic 0.52%
evaluation inside improvement	3.5	with a network, no guarantee; the critic must run 400 times faster
an answer key, first	3.1, 3.6	the protocol: where the grade is taken, and medians over seeds

- ▶ Deck 3.2 also found one thing the plan did not expect: **rolling the demo out further made it worse**, at every step size tried.
- ▶ **Six debts, six prices, one unit. What is still owed is an equilibrium, a government, and a test that is not a residual.**

Source: Decks 3.3–3.5, 7×2 grid unless stated; the continuous demo from Session 3 lab, block A2.

This Was an Illustration

Built in Session 3

an answer key before any learner
actor and critic, model known
pathwise gradients
a grade in units of consumption
small residual, not a right answer

Used in Session 4

the stationary equilibrium by classical methods first
household policy and value, distribution as an input
only the government differentiates through $\mathbf{x}'$
consumption-equivalent welfare across tax regimes
judge a solution by its simulated aggregates

- ▶ **Every number this session was on a growth model chosen so that an answer key exists** — a printable grid, a 400-point dynamic programme, a closed form. The point was to watch each idea get *graded*; none of it is a finding about an economy.
- ▶ **Session 4 starts on the problems these tools are for: Aiyagari, Krusell–Smith, transitions, taxation — where no answer key exists and a solution is judged by its simulated aggregates.**

Source: Miao & Feng, *AI for Economists*, ch. 12, “Knowing whether the answer is right”.